Tide retrospective: unconditional Tide 13 via discharge of harmonic `GibbsObservable` hypotheses (U2)

Daniel Murfet

7 May 2026

Abstract

This retrospective records an 80-line discharge tide. Tide 13 had proved that for the harmonic Gibbs measure with potential $L = (\lambda/2)x^2$, the cross-susceptibility derivative $\partial_h \mathrm{Cov}_h(x,x)\big|_{h=0}$ vanishes — but only conditionally on three caller-supplied analytic hypotheses for the observables $\varphi \in \{x,\ x\cdot x,\ x\cdot x\cdot x\}$. G4 had already proved a generic instance for the pow form x^k. This tide bridges the syntactic gap between the two forms via three function-extensional wrappers, and stitches them into Tide 13 to land an unconditional headline. The mathematical content is zero; the load-bearing observation is that the survey's size estimate (100–150 lines) was off by 2x because it papered over an unrecognised syntactic-form mismatch.

1 Setting

Tide 13 (cross-susceptibility derivative for harmonic, 6 May) closed the headline

$$\partial_h \, \mathrm{Cov}_h(x,x)\big|_{h=0} \;=\; 0$$

for the harmonic Gibbs measure $L_\lambda(x) = (\lambda/2)x^2$ with perturbation $A = \mathrm{id}$, *conditionally on three `Threepoint.GibbsObservable` hypotheses*: one each for the observables $\varphi \in \{x,\ x\cdot x,\ x\cdot x\cdot x\}$. The Tide 13 retrospective named the discharge of those three hypotheses as a follow-up tide; the May 7 surveys (v3, v4, v5) carried it as candidate O2 / U2.

G4 (harmonic-monomial GibbsObservable instances, 7 May, earlier in the day) had proved a generic theorem[1] giving, for any $k : \mathbb{N}$, a `GibbsObservable` instance for the observable $w \mapsto w^k$ at the harmonic potential with $A = \mathrm{id}$. At the time of G4 we believed this would close Tide 13's three hypotheses directly; closer inspection revealed a hidden obstacle. The hypotheses Tide 13 actually requires are written in *multiplicative* form $w \mapsto w \cdot w \cdot w$, not in `Pow.pow` form $w \mapsto w^k$. These are propositionally equal but not definitionally equal — Lean's = between `fun x ↦ x^2` and `fun x ↦ x*x` requires `funext x; ring`.

This tide bridges the gap and lands the unconditional version of Tide 13.

2 The thing formalised

Four new public theorems.

[1] The exported name is `Threepoint.harmonic_id_gibbsObservable_pow`.

Three syntactic-form wrappers in the harmonic-monomial GibbsObservable file (extending G4's file). For each $k \in \{1, 2, 3\}$,

$$\texttt{Threepoint.harmonic\_id\_gibbsObservable\_*} : \ \texttt{GibbsObservable}\ (\text{volume})\ L_\lambda\ \text{id}\ t\ \varphi_k$$

where $(\varphi_1, \varphi_2, \varphi_3) = (\texttt{fun x} \mapsto \texttt{x},\ \texttt{fun x} \mapsto \texttt{x*x},\ \texttt{fun x} \mapsto \texttt{x*x*x})$. Suffixes `_id`, `_mul_self`, `_mul_mul_self` indicate the multiplicative form of the observable.

Unconditional Tide 13 headline [2] For all $\lambda, t > 0$,

$$\texttt{HasDerivAt}\ \ (h \mapsto \mathrm{Cov}_h(x, x))\ \ 0\ \ 0.$$

The hypothesis bundle of Tide 13 has been folded into the applicability conditions of G4's `pow` instance, which require only $0 < \lambda$ and $0 < t$.

3 Proof strategy

Each wrapper proof is three lines: apply G4 at the relevant k; prove the function-extensional equality $\texttt{fun}\ x \mapsto x^k = \texttt{fun}\ x \mapsto \cdots$ by `funext; ring`; rewrite-into and `exact`.

```
theorem harmonic_id_gibbsObservable_mul_self
  (hlam : 0 < lam) (ht : 0 < t) :
  GibbsObservable volume L_lam id t (fun x ⇒ x * x) := by
    have h := harmonic_id_gibbsObservable_pow hlam ht 2
    have heq : (fun x : ℝ ⇒ x ^ 2) = (fun x ⇒ x * x) := by
        funext x; ring
    rwa [heq] at h
```

The unconditional headline composes the three wrappers with Tide 13's conditional theorem in one line:

```
exact cov_h_id_id_deriv_harmonic_eq_zero hlam ht
  (harmonic_id_gibbsObservable_id hlam ht)
  (harmonic_id_gibbsObservable_mul_self hlam ht)
  (harmonic_id_gibbsObservable_mul_mul_self hlam ht)
```

The deepest tactic in the entire tide is `ring`.

4 Roadblocks and resolutions

There were no roadblocks. The first attempt built clean modulo a missing import line in the cross-susceptibility file (fixed by importing the monomial-instances module).

GPT-5.5 Pro's deliberation, returned in parallel with the formalisation, suggested an *A-minus* variant: drop the named wrappers and inline the discharges as `by simpa [...] using ...` arguments to the conditional theorem. The trade-off is line count (about 30 fewer) versus discoverable API. We landed Candidate A — three named wrappers — on the bet that the discoverability cost is small and that future tides reaching for harmonic-id GibbsObservable instances at the $\{x,\ x \cdot x,\ x \cdot x \cdot x\}$ slot would prefer named references over a custom `simpa` incantation each time. If the second consumer turns out to want only the headline, the A-to-A-minus refactor is mechanical.

[2] Exported as `cov_h_id_id_deriv_harmonic_eq_zero_unconditional` in the cross-susceptibility derivative file.

5 What was Mathlib, what was new

Mathlib. `funext` (function extensionality introduction), `ring` (commutative-ring normalisation), and `rwa` (rewrite-then-exact). All three appear in dozens of laplace files; nothing new from Mathlib.

Laplace seabed (existing). G4's `Threepoint.harmonic_id_gibbsObservable_pow` (the load-bearing analytic content) and Tide 13's `cov_h_id_id_deriv_harmonic_eq_zero` (the conditional headline that the new theorem closes).

New. Three syntactic-form wrappers and one composition theorem. Total: four public theorems, 80 lines, of which the proof bodies are about 16 lines and the rest is documentation.

6 Lessons

Survey size estimates can be off by 2x when a syntactic-form mismatch is hidden inside the candidate description. The v5 survey carried U2 as "˜150 lines (possibly less post-O1-audit)"; the v4 and v3 surveys had similar estimates. The language used ("mixed-power" `GibbsObservable`) suggested a genuinely new analytic content beyond G4 — the user's prompt explicitly described "six remaining hypotheses out of the nine that Tide 13 left conditional, after G4 closed the three pure-monomial cases." Inspection of Tide 13's actual signature showed three hypotheses, not nine; and the obstacle was syntactic, not analytic. The lesson is procedural: when the tide picker is summarising a candidate from a retrospective Follow-ups section, surface the *Lean-level signature mismatch* explicitly if there is one, even when it would be obvious to anyone who opens the file. Otherwise the narrative "mixed-power" description carries forward through three survey revisions and overestimates the cost.

Inline `simpa` versus named wrappers is a small choice deferred to user style. GPT's A-minus suggestion is genuinely shorter; landing the named wrappers (Candidate A) is genuinely more discoverable for future consumers. Auto-mode picked A; the deletion back to A-minus is mechanical if needed. Worth recording as a decision-class rather than a single-call style.

The `ring` tactic suffices for all $x^k = x \cdot x \cdot \cdots \cdot x$ normalisations in this regime. For small k the conventional `ring` call closes the `funext`'d goal directly without needing simp hints. For larger k (perhaps 5+, by the time `ring`'s normalisation cost becomes visible), an explicit `simp` chain may be more efficient.[3]

7 Follow-ups

- *Anharmonic analogues.* The same syntactic-form bridge for the anharmonic Gibbs measure would discharge Tide 9 / G2 / G2+-style `GibbsObservable` hypotheses if they ever surface in the multiplicative form. Currently the anharmonic `kappa3`-asymptotic theorems don't expose `GibbsObservable` hypotheses at the headline level (Tide 9 uses unconditional integrability witnesses instead), so this is blocked on a use-case rather than a missing theorem.

[3] Hints to consider: `pow_one`, `pow_two`, `pow_succ`.

- *Higher-power discharges.* A `harmonic_id_gibbsObservable_pow_k` for $k = 4, 5, 6$ multiplicative forms is a direct re-application of the same template, $\sim$10 lines per case. Defer until a consumer (e.g. a κ_4 tide) reaches for them.

- *Generic `Threepoint.GibbsObservable.congr_fun` transport lemma.* If a second consumer (say, the anharmonic case above) wants the same syntactic-form transport, lift to a single generic transport lemma and rewrite the three wrappers as one-line corollaries. GPT-5.5 Pro flagged this as a "add when 2nd consumer appears" candidate; right call.

- *Update primer / threepoint paper for the unconditional headline.* The primer's discussion of cross-susceptibility derivatives now has an unconditional Lean reference; pin `\leanref` for both papers in the next process tide. Pairs with K2 in the survey.

- *2D / multivariate analogues.* The same discharge pattern would land in a multi-index harmonic file once a multi-index `GibbsObservable` instance theorem exists. Currently no such theorem; blocked.